



Characterization of water residence time, nutrients, phytoplankton and related water quality constituents in the Cache Slough Complex of the California Sacramento-San Joaquin Delta in 2017 and 2018

Methods

This study involved characterizing nutrients, phytoplankton, and related water quality constituents along with water residence times in the Cache Slough Complex of the north San Francisco Bay Delta using high-resolution mapping and discrete sample collection. Data collection took place on three separate days: October 11, 2017; May 30, 2018; and October 16, 2018. The main purpose of this study was to observe changes in water quality parameters and relate those changes to water residence time to gain insight into rates of nutrient transformation.

Sampling and analysis

Underway high resolution (HR) water sampling and onboard analyses:

Surveys were conducted using the USGS R/V Mary Landsteiner following the approach described by Bergamaschi et al. (2020). On each day data were collected during daylight hours. For analysis, sample water was continuously pumped onto the boat using two SHURflo Aqua King II pumps at a pressure of 55 psi and rate of 5 GPM (SHURflo Pentair, London, United Kingdom) while underway at speeds up to 30 mph (13 m/s) using a pick-up tube mounted at a fixed depth of approximately 1 m below the surface, routed through a 178 μ m in-line strainer (Cole Parmer; EW-29595-47, Vernon Hills, IL) to remove large debris and into a pressure-compensated manifold that maintained system pressure at a proscribed

level irrespective of boat speed (Downing et al., 2016). A 2-stage debubbler was used to remove bubbles that can interfere with optical measurements. Sample water was split into separate flow paths for each analytical flow path, with constant flow rates maintained by metering the discharge line, except for the flow path serving the open split interface, which was metered on the feed line. Flows were continuously monitored using a sight gauge for each flow path.

The manifold delivered water to the following flowpaths: (1) A flow-through system consisting of a thermosalinograph that recorded temperature and conductance (Sea-Bird Scientific SB45 (TSG), Bellevue, WA); fluorometers that measured fluorescence of chlorophyll (fCHL) and fluorescence of dissolved organic matter (fDOM, WETLabs WETstar (WS), Philomath, OR); a beam transmissometer that recorded transmittance and attenuation (WETLabs model C-Star transmissometer (CStar), Philomath, OR), a nitrate (NO_3) analyzer (SUNA V2; Sea-Bird Scientific, Bellevue, WA). (2) A flow chamber for a multiparameter water quality sonde (YSI EXO2; Xylem Inc. (EXO), Rye Brook, New York) equipped with sensors to measure temperature, specific conductance (SpC), turbidity, pH, dissolved oxygen (DO), fDOM and fCHL that then passed water to a fluorometer designed to measure different algal classes (FluoroProbe III (FP); BBE Moldaenke, Kiel, Germany). (3) An open-split interface at atmospheric pressure that served water filtered through a 0.45 micron high-capacity in-line groundwater sampling capsule filter (PALL Corporation, Port Washington, New York) to the on-board ammonium (NH_4) analyzer water isotope ($\delta^2\text{H}$, $\delta^{18}\text{O}$) analyzer.

The NH_4 analyzer was a continuous flow, gas diffusion/conductivity-based (Carlson 1986) analyzer for ammonium (NH_4) analysis (TL-2800; Timberline Instruments, Boulder, CO) that was modified for field operation and continuous data collection by the manufacturer. Modifications included installation in a ruggedized housing, addition of an automated line-switching valve, addition of a heating unit to maintain the instrument at a constant above-ambient temperature, and changes to the software. Full standard curves were run at the beginning and end of each day and partial curves run throughout the day each time the boat stopped to sample. While underway, the analyzer was run in continuous mode with frequent periodic introduction of blank solutions to track instrument drift. The NH_4 analyzer was connected to a stand-alone computer and collected data through its native software.

The water isotope analyzer employs cavity ring down spectroscopy to measure isotopic concentrations of $\delta^2\text{H}$ and $\delta^{18}\text{O}$ in real-time (Picarro Inc., Sunnyvale, CA). The instrument is a continuous flow instrument comprised of 2 separate instruments. The continuous water analyzer (CWS) pumps sample water from the open-split interface and vaporizes samples water across a membrane. The CWS delivers vaporized water to the CRDS where water isotope concentrations are measured and reported in real-time (Downing et al. 2016). Tertiary standards (Kona water) are measured approximately every 4 hours and are used to track instrument drift throughout the day. The instrument has an onboard computer where data is collected and stored through its native software.

All instrumentation was cleaned, and calibrations were checked prior to each use following the manufacturer's recommendation or as described below. Data for most instruments was recorded at 1-second frequency on a single data logger (CR6, Campbell Scientific, Logan, Utah, United States) together with a timestamp and boat position obtained from a high-resolution global positioning system receiver (16X-HVS, Garmin,

Olathe, Kansas). The FluoroProbe logged data internally and to the host software. All data were displayed in real time, so scientists on board could respond when they observed changes relevant to study objectives, or if issues with flow or instrument performance were noted.

Discrete water sampling (DS) and laboratory analyses:

Discrete water samples were collected either underway or—in the case of more extensive sample requirements—while stopped on station. Discrete samples were collected while underway either using a fourth flow path on the sampling manifold or using a separate system with a separate pick-up tube mounted adjacent to the main sampling pick-up and pumping sample through a 0.45-micron filter into a sample bottle. At designated sampling stations (approximately 10 per day) samples for determination of NH_4 , NO_3 , PO_4 , dissolved organic carbon (DOC), and total dissolved nitrogen (TDN) concentrations were collected using this system. Additionally, unfiltered samples were collected approximately from 1 m depth using a submersible pump for analysis of chlorophyll-*a* (Chl-*a*), pheophytin, phycocyanin, allophycocyanin, and phycoerythrin pigments, phytoenumeration, picocyanobacterial counts, and water isotopes. Samples for phytoplankton enumeration and picocyanobacteria counts were fixed immediately with Lugol's iodine solution (4 mL) and formaldehyde (1 mL), respectively. Sample splits for accessory pigment analysis were collected by USGS but processed and analyzed by California State University Maritime Academy.

All samples were stored on ice in the dark during transit to the laboratory where they were stored at 2°C. Dissolved organic carbon samples were preserved to <2 pH with high-purity sulphuric acid the day following collection and stored at 2°C. Samples for pigment analysis were filtered within 24 hours of collection and immediately frozen (-90°C). Samples for phycocyanin, allophycocyanin, and phycoerythrin pigment analysis were filtered through a 0.3 µm nominal pore size glass fiber filter (Advantec MFS, Inc, Dublin, CA). Chilled nutrient samples were shipped to the USGS National Water Quality Laboratory (Denver, CO). Frozen filters for analysis of phycocyanin, allophycocyanin, and phycoerythrin were shipped to BSA Environmental Services, Inc. (Beachwood, OH) as were preserved picocyanobacteria samples. Water isotope samples were sent to the U.S. Geological Survey Stable Isotope Laboratory (Reston, VA).

Concentrations of nutrients were determined colorimetrically using an automated-segmented flow analyzer. Concentrations of nitrogen as nitrite ($\text{NO}_2\text{-N}$) and as nitrate plus nitrite ($\text{NO}_3\text{-N} + \text{NO}_2\text{-N}$) were determined by colorimetric analysis (Fishman, 1993; Patton and Kryskalla, 2011), ammonium as nitrogen ($\text{NH}_4\text{-N}$) was determined by colorimetric analysis after reaction with salicylate-hypochlorite (Fishman, 1993), and total dissolved nitrogen (TDN) was determined by alkaline persulfate digestion (Patton and Kryskalla 2003). Orthophosphate as phosphorous ($\text{PO}_4\text{-P}$; also referred to as soluble reactive phosphate) was determined by colorimetric analysis after reaction with NH_4 molybdate and reduction with ascorbic acid (Patton and Truitt, 1992).

Samples for phytoplankton pigments were collected on Whatman® glass fiber (GF/F, nominal pore size 0.7-µm diameter) filters and Nuclepore® filters with 5 µm pore size to capture the larger cells. Chlorophyll-*a* concentration (Nuclepore® filters) on the Nuclepore® filters, together with the GF/F filters, was used to obtain chlorophyll-*a* in the > 5 µm and < 5µm cell size fractions. Prior to analysis, filters were stored frozen at -20° C

for up to one week before being analyzed at Cal Maritime or, with a subset of GF/F filters, shipped to an outside laboratory for quantification of accessory pigments diagnostic of phytoplankton functional groups (see below). Nuclepore® filters were analyzed on a Turner Designs Model 10-AU fluorometer after a 24 h pigment extraction in 90% v/v acetone at 4 °C (Arar and Collins 1997) and correcting for phaeophytin with a 10% hydrochloric acid addition. The fluorometer was calibrated with a commercial Chlorophyll-*a* standard (Turner Designs)

Dissolved organic carbon concentrations were measured using a total organic carbon analyzer (TOC-V_{CSH}, Shimadzu Scientific Instruments, Columbia, Maryland) using high-temperature catalytic combustion according to a modified version of USEPA method 415.3 (Potter and Wimsatt, 2012). Absorbance spectra and fluorescence EEMs were simultaneously measured using an Aqualog® equipped with a photodiode and charge-coupled device (CCD) (Horiba Instruments, NJ, USA) according to the method described by Hansen et al. (2018). Briefly, excitation and absorbance spectra were measured using a 150 W xenon lamp, a 5 nm bandpass, and a 1 s integration time at wavelengths of 240–600 nm. Emission spectra were collected with a CCD at approximately 1.64 nm (4 pixel) intervals at wavelengths of 250–600 nm. To reduce ultraviolet light exposure of the sample and limit the effects of photobleaching, excitation and absorbance wavelengths were scanned from low to high energy (i.e., visible to ultraviolet range) during analysis. Peaks, ratios, and slopes reported in the data were calculated as described by Hansen et al. (2018).

Phycocyanin, allophycocyanin, and phycoerythrin were determined by azolectin-CHAPS buffer extraction as described by (Zimba, 2012) and modified from (Bennett and Bogorad, 1973) at BSA Environmental Services. Phytoplankton enumeration was completed in accordance with the American Public Health Association (APHA) Standard Method 10200 (2012). Picocyanobacteria direct counts were enumerated via epifluorescence microscopy as described by (Murrell and Lores, 2004). Phycoerythrin-containing cells (PE) and phycocyanin-containing cells (PC) were counted separately and then summed to obtain total picocyanobacteria densities for cells sized between 0.2 and 2.0 microns.

Accessory pigment samples were processed by HPLC (High Performance Liquid Chromatograph, Agilent) at the Horn Point Analytical Laboratory at the University of Maryland, following the protocols of Van Heukelem and Thomas (2001). Major accessory pigments obtained from HPLC are reflective of the algal functional group. These pigments differ slightly from the diagnostic pigments of the FluoroProbe since the HPLC cannot analyze water-soluble pigments such as phycocyanin or phycoerythrin. The HPLC diagnostic pigments are chlorophyll-b for the chlorophytes (green algae), fucoxanthin (brown pigment diagnostic of the diatoms), alloxanthin for the cryptophytes, zeaxanthin for the cyanobacteria and peridinin for the dinoflagellates. Peridinin was not observed in HPLC results and not included in this report.

Analysis of Stable Isotope Parameters

$\delta^2\text{H}$ and $\delta^{18}\text{O}$ of H_2O : Sample were analyzed at USGS Reston Stable Isotope Lab using dual-inlet isotope ratio mass spectrometry according to the methods of Révész and Coplen (2008a&b).

$\delta^{13}\text{C}$, $\delta^{15}\text{N}$, and C:N ratio of Particulate Organic Matter: Particulate organic matter (POM) was collected via filtration through 0.7 μm pre-combusted glass fiber filters, prepared, and analyzed at MPSIL following the method described in Kendall et al. (2001), using an Optima mass spectrometer. Samples are weighed in silver boats and then vapor acidified to remove any carbonate prior to analysis. C:N values are reported as atomic (at) ratios.

$\delta^{13}\text{C}$, $\delta^{15}\text{N}$, $\delta^{34}\text{S}$, C:N ratio, and C:S ratio of Dissolved Organic Matter: Dissolved organic matter (DOM) was isolated by passing sample water filtered to 0.2 μm through PPL resin columns. The dissolved organic matter was then eluted from the column using methanol and collected in tin capsules for analysis at MPSIL using a Vario Elemental Analyzer coupled to an IsoPrime mass spectrometer. C:N and C:S values are reported as atomic (at) ratios.

$\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ of Nitrate: Filtered water samples were analyzed at UC-Davis Stable Isotope Facility using the bacterial denitrification method of Sigman et al. (2001) and Casciotti et al. (2002). NO_2^- was removed from samples prior to analysis using the method of Granger and Sigman (2009).

$\delta^{15}\text{N}$ of Ammonium: Filtered water samples were analyzed at MPSIL using a minor modification of the Holmes et al. (1998) microdiffusion method using an elemental analyzer connected to an Optima or IsoPrime mass spectrometer.

The isotope data are reported in permil (‰) notation relative to the following international standards: AIR for $\delta^{15}\text{N}$, VSMOW for $\delta^{18}\text{O}$ and $\delta^2\text{H}$, VPDB for $\delta^{13}\text{C}$, and CDT for $\delta^{34}\text{S}$

Data processing

Data from onboard (in situ) continuous instruments not directly logged to the flow-through data collection system were merged based on timestamp to the nearest second. All data directly logged to the flow-through data collection system were processed using pandas software library (0.220) in Python (3.7.3) to remove periods of compromised data (e.g., flow blockages, bubbles) and to apply instrument corrections and unit conversions. A centered 20-second median filter was applied to all continuous flow through data to remove erroneous data (e.g., bubble or electrical noise), with the exception of the NH_4 data, in Python. Final high-resolution nitrate and water isotope values were obtained by regressing in situ instrument response against concentrations obtained from laboratory measurements of discrete samples collected through the course of the day. Individual sample results with more than three standard deviations from the regression were judged to be outliers (see below) and removed from the regression. One regression relationship for the entire study was developed for nitrate and for water isotope data a different relationship was developed for each study date resulting in 3 regression equations. The continuous, in situ ammonium data was processed by first correcting the voltage output of the instrument for baseline drift based on the instrument response during periodic measurements of organic-free deionized water and then using a regression model of the voltage response to standard concentrations run

throughout the day. A regression coefficient of $>0.97 R^2$ or better was judged to be acceptable. The ammonium data timestamp was corrected to account for the time lag associated with the microfluidic processing prior to integration with other data sources.

Larger differences between SUNA and laboratory nitrate concentration data are assumed to arise from 1) actual spatial heterogeneity in environmental nitrate concentrations between water entering the flow through system and that captured in the sample bottle, 2) interferences of turbidity and dissolved organic matter using the UV absorption method for nitrate, or 3) errors in sample handling and laboratory analysis. Excepting these potential sources of error, the field and laboratory nitrate measurements are typically well correlated with a linear fit. For the simple linear regressions, the SUNA NO₃ concentration data was the explanatory variable to compute the response variable of the laboratory NO₃ concentration (Bergamaschi et al., in press). P-values were <0.0001 for all regressions. Summary statistics and regressions for individual days are identified in Table 1.

Table 1. Summary statistics and linear regressions for nitrate (NO₃) concentration data obtained from high-resolution SUNA instrument and discrete samples analyzed using standard laboratory methods [n, number of observations; μM , micromolar; mean, mean high-resolution nitrate concentration; median, median high-resolution nitrate concentration; min, minimum high-resolution nitrate concentration; max, maximum high-resolution nitrate concentration; SS, sum of squared error; SSE, sum of squared estimate of error; MSE, mean square error; RMSE, root mean square error; %, percent; MSPE, model standard percentage error; R², coefficient of determination; *, asterisk indicates adjusted R²].

Parameter	n	min (μM)	max (μM)	median (μM)	mean (μM)	SS (μM)	SSE (μM)	MSE (μM)	RMSE (μM)	MSPE (%)	regression equation	p-value	R ²
NO ₃ (μM) (SUNA) HR	36	3.9	26.5	12.7	14.4	1274	46	1.3	2.2	21	$y = 0.9739$ (uncorrected NO ₃ (μM) (SUNA) HR) - 3.7003	<0.001	0.9082

Dissolved inorganic nitrogen (DIN) concentration was calculated as the sum of NO₃, NO₂ and NH₄—both for the continuous data and the discrete sample data. Laboratory DON concentration was calculated by subtracting NH₄, NO₃, and NO₂ data from the TDN data (i.e. DON = TDN – DIN). Total dissolved nitrogen (TDN) was calculated as the sum of DIN (NO₃ + NH₄) and DON. Continuous DOC and DON concentrations were developed by regressing onboard fDOM data against the measured DOC concentrations and calculated DON concentrations in discrete samples was determined according to modified methods of Snyder et al. (2018). Data exceeding 3 times the standard deviation of the residuals for DOC and 2 times the standard deviation of the residuals for DON were judged to be an outlier and not included in the model. The linear regression for estimated DOC = $0.072(\text{fDOM (QSU) (WS) HR}) + 0.351$ ($R^2 = 0.967$). The linear regression for estimated DON = $0.072(\text{fDOM (QSU) (WS) HR}) + 0.351$ ($R^2 = 0.67$).

Water residence time (WRT, τ) was calculated from the stable isotope of water data according to methods described in Downing et al. (2016). To calculate water residence time (WRT, τ), we had to first calculate the evaporation-to-inflow (E:I) ratio (Brooks et al. 2014; Gibson and Reid 2014). This ratio can be modeled using the Hydrocalculator software as follows:

$$X = E/I = (\delta I - \delta Q) / (\delta E - \delta Q)$$

Where δI is inflow water estimated as intersection of local evaporation line (LEL) and local meteoric water line (LMWL); δQ is outflowing water represented by field measured water isotope values; δE is evaporating water; and δA is isotopic values of ambient air, estimated using 'online isotopes in precipitation calculator' (OIPC) algorithm (Bowen and Revenaugh 2003). The Hydrocalculator solves the above equation for X for each data point (Skrzypek et al. 2015). The model also requires atmospheric temperature and relative humidity, obtained from the California Irrigation Management Information System (CIMIS) network. The LMWL was developed from isotope data collected within the San Francisco Bay-Delta and Sacramento River over the last decade (Craig 1961; Kendall et al. 2015, 2001). The LEL was produced from the field data (Craig 1961). E:I values are then used to calculate WRT as follows (Brooks et al. 2014; Gibson and Reid 2014): $\tau = x \cdot v / E$ Where X is the E:I ratio, V is the average depth of the water column (cm), and E is the evapotranspiration per day (ETo) in cm d⁻¹ (Jensen 2010). ETo values were obtained from CIMIS.

Spatial Alignment (SA)

Data was spatially aligned to a common spatial framework to facilitate comparisons between dates and to minimize the effects of differential data density on the spatial interpolation calculations. The spatial alignment was performed using the geopandas software library in python, which assigned median-filtered data points to polygon shapefiles that are spaced at approximately 150 m intervals. Spatially aligned data was then aggregated by polygon shapefile to obtain summary statistics of all median-filtered data within a polygon.

Interpolated water quality maps were created from the spatially aligned data using ArcGIS Pro Spline with Barriers tool (Terzopoulos and Witkin, 1988). The raster output was smoothed with the Focal Statistics tool, and interpolated values were extracted to the spatially aligned points to create continuous spatially aligned data across the mapping domain.

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